AI LEGAL DOCUMENT SUMMERIZER

PROJECT WORK
PHASE-I REPORT

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ABSTRACT

The Legal Document Summarizer is an AI-powered system designed to automate the extraction and summarization of complex legal documents using Natural Language Processing (NLP) and deep learning methodologies. This project addresses the critical challenges faced in manual legal document review, including high operational costs, significant time delays, and potential human errors. The system employs transformer-based models fine-tuned on legal corpora to generate both extractive and abstractive summaries with precise clause-level citations.

By integrating Named Entity Recognition (NER) for key information extraction and implementing full traceability through hyperlinked source references, the system transforms legal workflows by reducing manual review time by 50-70% while maintaining accuracy and auditability.

The solution is delivered through a secure web application with a modular architecture comprising a responsive frontend (React.js), RESTful backend API (Flask), and specialized AI engine powered by Hugging Face transformers. This document presents the comprehensive system design, literature survey, technical specifications, and implementation strategy for developing a domain-specific legal summarization platform that ensures quick comprehension, verified summaries, and broad document support across diverse legal document types.

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LISTOFABBREVIATIONS

AI - Artificial Intelligence
NLP - Natural Language Processing
NER - Named Entity Recognition
OCR - Optical Character Recognition
API - Application Programming Interface
REST - Representational State Transfer
DFD - Data Flow Diagram
UI - User Interface
PDF - Portable Document Format
DOCX - Microsoft Word Document Format
TLS - Transport Layer Security
SSL - Secure Sockets Layer
CI/CD - Continuous Integration/Continuous Deployment
BART - Bidirectional and Auto-Regressive Transformers
T5 - Text-to-Text Transfer Transformer
ROUGE - Recall-Oriented Understudy for Gisting Evaluation
F1 - F1 Score (Harmonic Mean of Precision and Recall)
UAT - User Acceptance Testing
RAM - Random Access Memory
ML - Machine Learning
IDE - Integrated Development Environment
CSS - Cascading Style Sheets